Dense Retrieval, Semantic Similarity, and Proxy Substitution

Why the Most Similar Passage Is Not Necessarily the Best Evidence

AI Memory and Reality Drift - Note #3 v1

A. Jacobs - Semantic Fidelity Project

The Basic Pattern

Dense retrieval represents queries and passages as vectors and selects items that are close in embedding space. This makes it possible to retrieve conceptually related language even when the query and source share few exact words.

The method is powerful because semantic similarity often correlates with relevance. But relevance is broader than similarity. The best evidence may be less lexically or conceptually similar because it supplies a counterexample, controlling definition, current version, causal explanation, or authoritative contradiction.

The architectural substitution is simple: a difficult judgment about evidentiary relevance becomes a computable distance between representations.

When Similarity Becomes Relevance

A retriever needs a score that can rank millions of candidates quickly. Embedding similarity supplies that score. Top-k retrieval then turns a continuous geometry into a small evidence set.

Once this pipeline is evaluated and tuned using retrieval benchmarks, teams improve the system's ability to return passages labeled relevant. The representation of relevance becomes the optimization surface. Passages that satisfy the scoring function gain access to the context window, while other forms of evidence remain causally absent.

How Proxy Substitution Emerges

Stage 1 - Representation. Queries and passages are compressed into vectors.

Stage 2 - Similarity scoring. Distance is used to estimate conceptual relevance.

Stage 3 - Ranking. The highest-scoring candidates are selected under a fixed context budget.

Stage 4 - Optimization. Embeddings, chunking, and queries are tuned to improve retrieval metrics.

Stage 5 - Substitution. Similarity becomes the operative definition of what counts as evidence.

The Optimization Trap and Reality Drift

The Optimization Trap occurs when a useful proxy becomes the target of improvement and gradually displaces the outcome it was meant to support. Dense retrieval does not need to become inaccurate for this to occur. It can become increasingly good at similarity while remaining uncertain about authority, time, causality, or contradiction.

Reality Drift describes the resulting system-level condition. The application continues answering questions and retrieval scores remain healthy, but the selected evidence can lose alignment with what a careful inquiry would require.

The Representation Stack

A source passage becomes a chunk. The chunk becomes an embedding. The query becomes another embedding. Their geometric relationship becomes a relevance score. The score becomes a ranking. The ranking becomes the model's temporary evidence environment.

By the time the answer is generated, similarity has traveled through the stack and acquired the appearance of evidentiary authority.

Examples Across AI Systems

Temporal questions: An obsolete passage can be nearly identical in meaning to the current answer and therefore rank highly.

Policy questions: A general policy may be more similar to the query than the narrow exception controlling the case.

Scientific questions: A popular supporting paper may be closer to the query than a methodologically stronger criticism.

Troubleshooting: A common failure mode may outrank a less similar but product-specific cause.

Conclusion

Dense retrieval is a mechanism for finding representational proximity. It becomes risky when proximity silently absorbs the broader meaning of relevance.

The solution is not abandoning embeddings. It is restoring additional constraints through sparse signals, metadata, source authority, time, diversity, contradiction search, and reranking. Similarity should open the inquiry, not define its result.

Related Phrases and Concepts

- dense retrieval
- semantic similarity
- vector search
- embedding space
- approximate nearest neighbors
- cosine similarity
- hybrid retrieval

Semantic Fidelity Project Resources

- Research Library (GitHub)
- Articles & Essays (Substack)
- Semantic Fidelity Glossary